



How to Configure NIC in VMware for EdgeOne

Application Note

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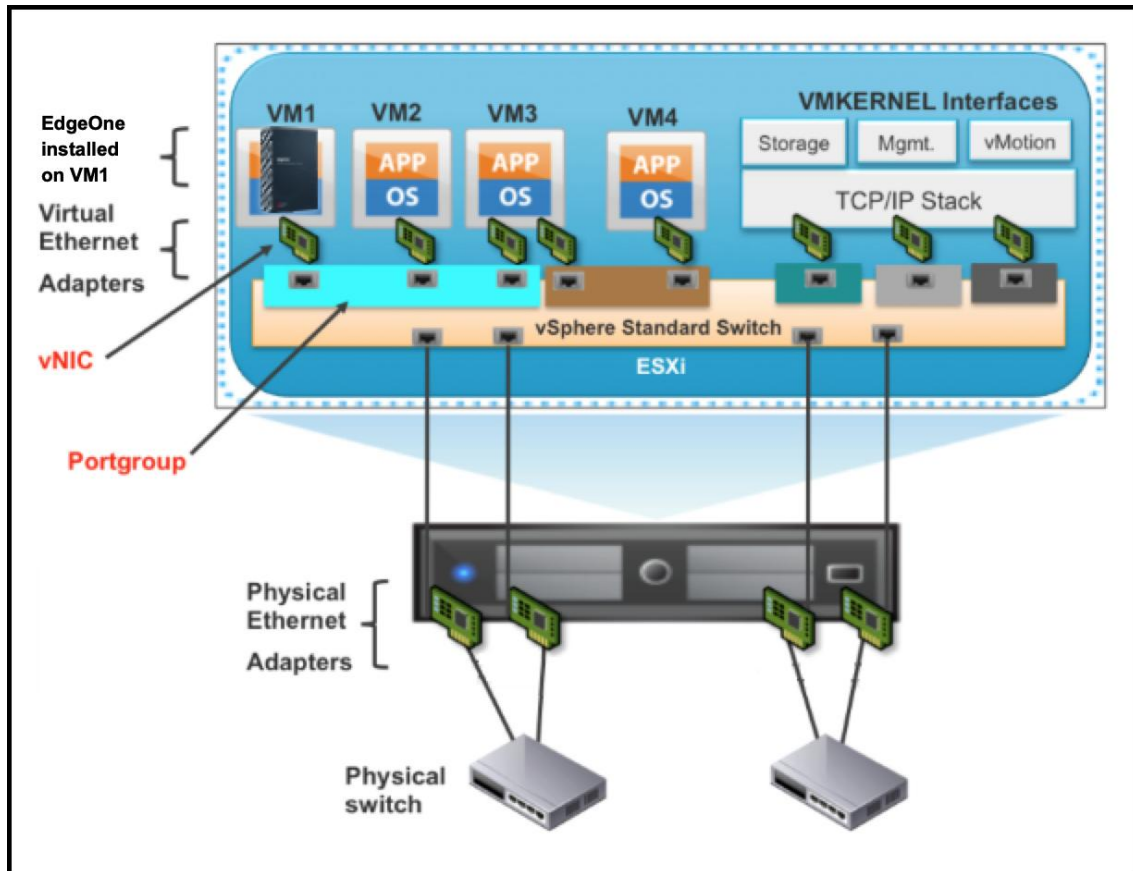
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1 Introduction



In complex deployment scenarios, it is common for system integrators to encounter environments where multiple network interface cards (NICs) are installed on a single hypervisor. This configuration typically arises in virtualized infrastructures where multiple virtual machines (VMs) or virtual network functions (VNFs) are hosted on the same physical hardware. Each VM may serve a different purpose—such as SCADA, HMI, or historian applications—and may require dedicated communication paths to different external services or host systems. To meet performance, security, or routing requirements, these virtual images are often assigned separate NICs to isolate network traffic by application or function.

In such cases, deploying the EdgeOne platform within this kind of virtualized environment requires special attention to how the NICs are configured. Ensuring that the correct NIC is assigned to EdgeOne is critical for maintaining proper connectivity with external networks.

This application note is intended to guide system integrators through the detailed steps required to configure EdgeOne for successful deployment in a hypervisor with multiple NICs. It provides practical instructions on how to:

- Identify which NIC is physically connected to the desired network segment.
- Assign and verify NIC usage at the hypervisor level.
- Perform network setup and troubleshooting inside the EdgeOne vshell environment.

By following this guide, integrators can ensure that EdgeOne is properly connected to its intended network environment, facilitating reliable data communication and platform operation.

2 Network Configuration Requirements

To establish a stable and functional connection for EdgeOne in this multi-NIC setup, the system integrator must first identify and configure the appropriate physical NIC from within the hypervisor's management interface. Depending on the hypervisor (such as VMware ESXi, Microsoft Hyper-V, or KVM), this process may involve:

- Assigning the correct physical NIC to the EdgeOne VM or virtual appliance.
- Ensuring that the NIC is properly bridged or mapped to the virtual network adapter used by EdgeOne.

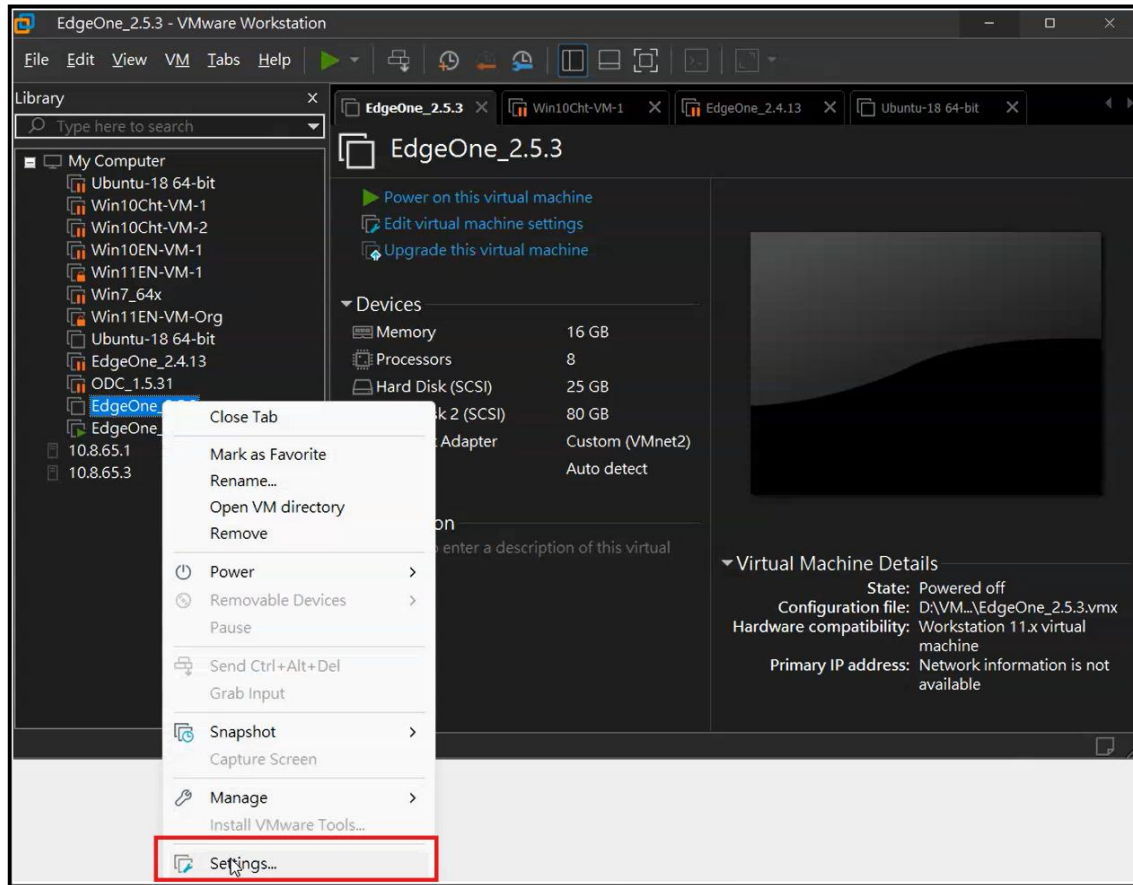
Once the hardware-level NIC assignment is completed, further configuration is required within the EdgeOne platform itself. This is done via the **vshell**, where the administrator can:

- Verify NIC detection and status.
- Set IP addressing (static or DHCP).
- Define default gateways and DNS servers.
- Perform connectivity tests (e.g., ping, traceroute) to validate external network access.

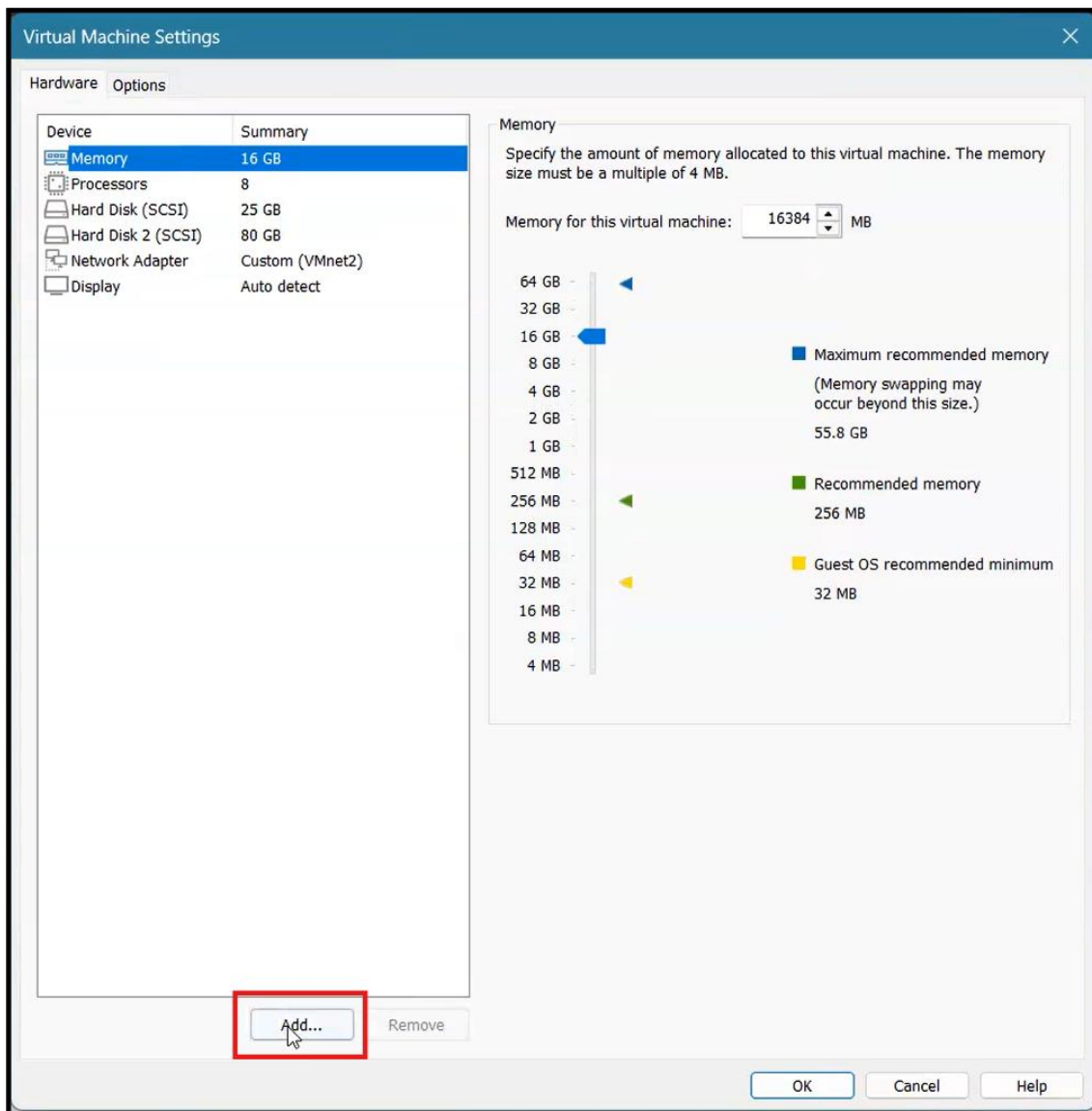
3 Configuring NIC for EdgeOne

Procedure

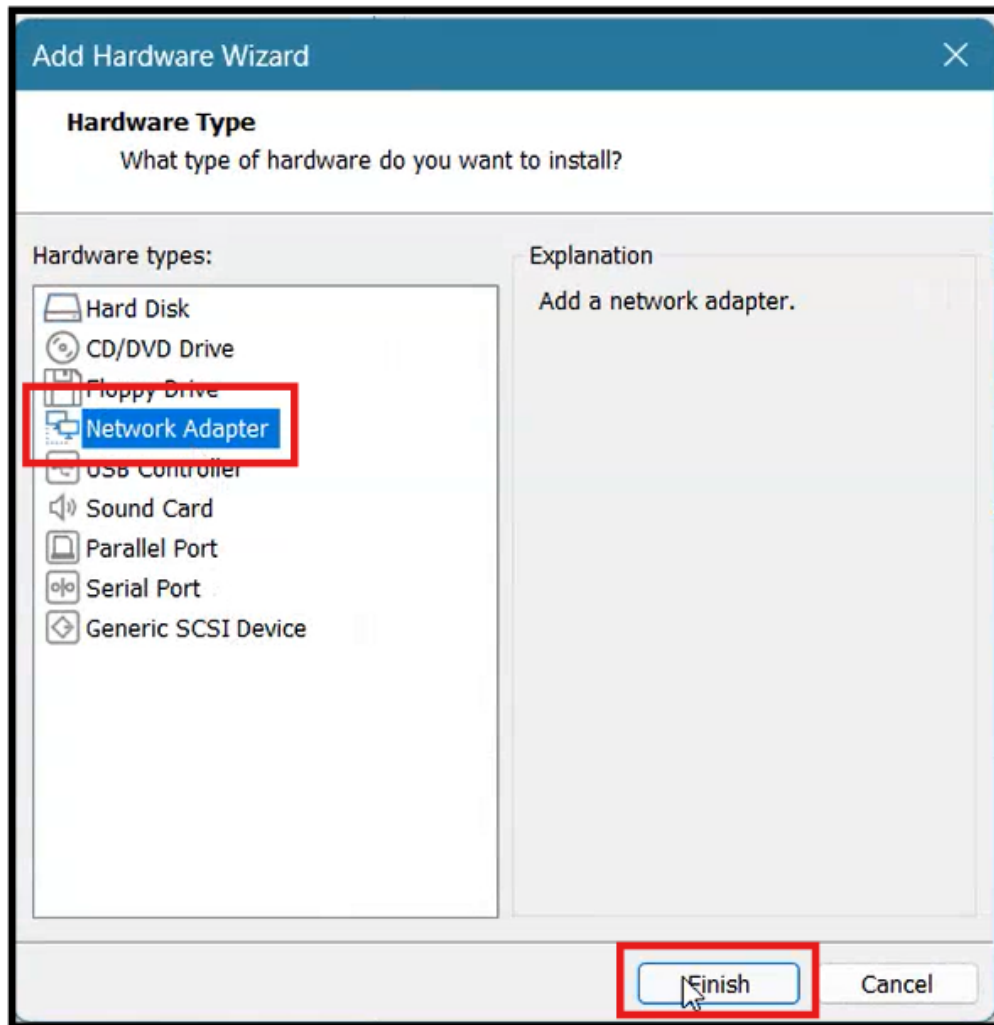
1. Open VMware, right-click your EdgeOne instance, and click [Settings] from the popup menu.



2. On the [Virtual Machine Settings] window, click [Add].



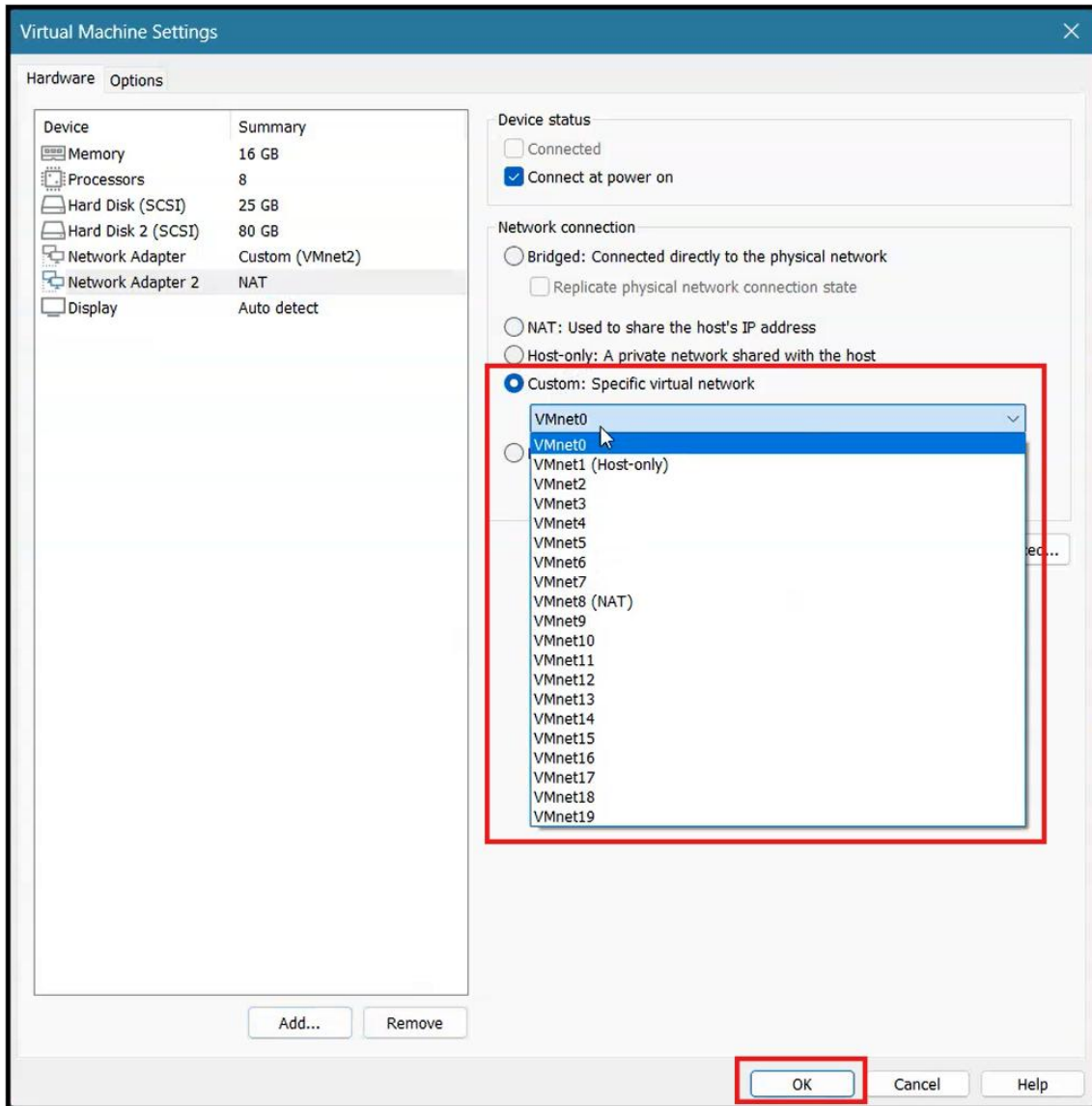
3. On the [Add Hardware Wizard] window, select [Network Adapter] and click [Finish].



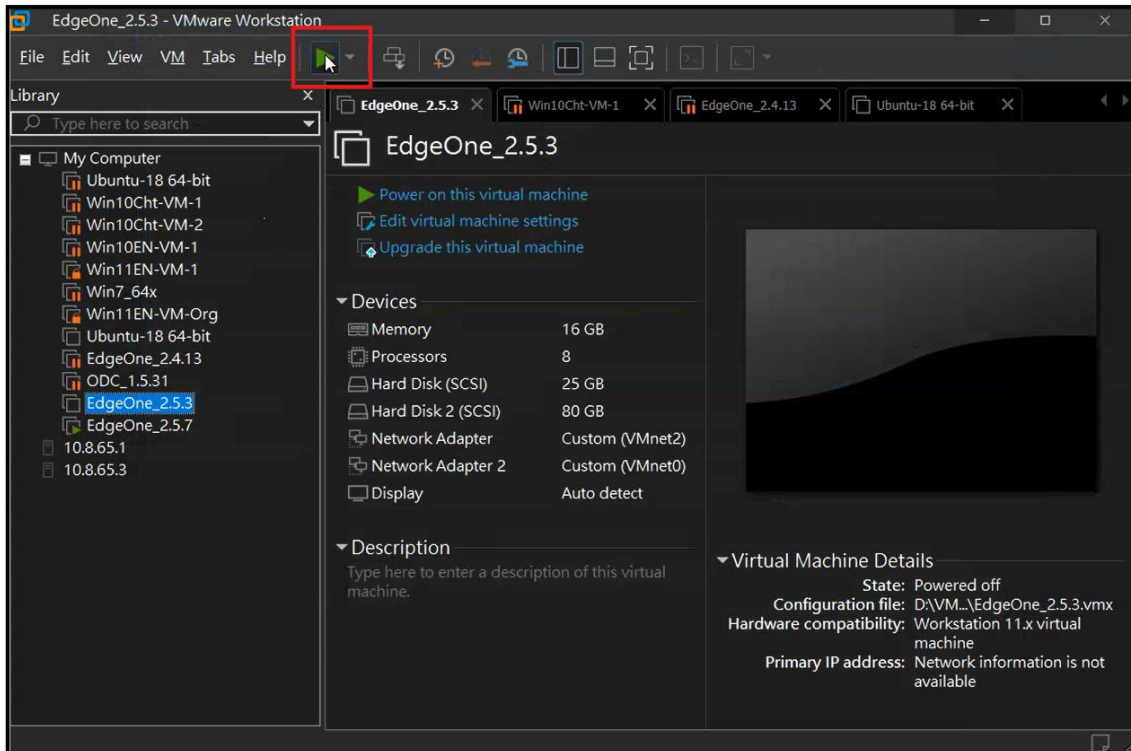
4. On the [Virtual Machine Settings] window, change the default network connection setting [NAT: Used to share the host's IP address] to [Custom: Specific virtual network], select one NIC from the drop-down list, and click [OK].



If the VMware host is equipped with multiple NICs and you do not specify which NIC to map to EdgeOne, EdgeOne will use the first NIC in the list by default.



5. Reboot the EdgeOne VM.



6. Log in to the EdgeOne CLI and type either of the following commands to add the new interface (eth1) for the EdgeOne instance.

```
iface add eth1 --method static --address X.X.X.X --gateway X.X.X.X --netmask X.X.X.X
```

or

```
iface add eth1 --method dhcp
```

 The placeholders for IP address, gateway, and netmask are user-defined.

7. Type the following command to activate the interface (eth1).

```
iface up eth1
```

8. Type the following command to view the information of the eth0 and eth1 interfaces.

```
iface ls
```

9. Type either of the following commands to configure route settings for the new interface (eth1). All traffic to IP address Y.Y.Y.Y or subnet Y.Y.Y.Y/Y will be routed to gateway X.X.X.X via the interface eth1.

```
iproute add eth1 --gateway X.X.X.X --nat Y.Y.Y.Y
```

or

```
iproute add eth1 --gateway X.X.X.X --nat Y.Y.Y.Y/Y
```

10. Type the following command to view the route settings.

```
iproute ls
```

11. Type the following command to make the route settings take effect.

```
iface restart eth1
```

 You do not need to reboot EdgeOne because it was already rebooted when you added the new interface.

4 Troubleshooting NIC Configuration Issues

Procedure

1. Use the following command to check if the default EdgeOne NIC (eth0) and the user-assigned NIC in VMware (eth1) are both assigned IP addresses from different subnets.

```
iface ls
```

2. If both NICs have IP addresses, use the following command to check if the gateways (X.X.X.X) for both NICs are accessible.

```
ping X.X.X.X
```

For example:

```
ping 192.168.10.254
```

```
ping 192.168.20.254
```

3. Alternatively, you can use the following command to check if both NICs are routable when the ping command fails.

```
dx traceroute X.X.X.X --interface Y
```

For example:

```
dx traceroute 192.168.10.254 --interface eth0
```

```
dx traceroute 192.168.20.254 --interface eth1
```

```

If you want to exit this shell, please type 'exit' or 'Ctrl-D'.
Reminder: The system will automatically logout idle users after 60 minutes.
$ iface ls
{
  {
    "Name": "lo",
    "Family": "inet",
    "Method": "loopback"
  },
  {
    "Name": "eth0",
    "Family": "inet",
    "Method": "dhcp"
  }
}
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
   link/ether 00:50:56:a4:0d:ae brd ff:ff:ff:ff:ff:ff
   altname enp2s0
   inet 10.8.67.222/18 brd 10.8.127.255 scope global dynamic eth0
       valid_lft 59496sec preferred_lft 59496sec

```

```

0.0.0.0) 4.258 ms
$ dx traceroute 8.8.8.8 --interface eth1
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1 10.8.127.254 (10.8.127.254) 0.900 ms 0.755 ms 0.699 ms
 2 * * *
 3 168-95-82-198.tpkt-3310.hinet.net (168.95.82.198) 2.676 ms 2.628 ms 2.685 ms
 4 * * *
 5 220-128-9-33.tylc-3032.hinet.net (220.128.9.33) 3.969 ms 220-128-9-37.tylc-3032.hinet.net (220.128.9.37) 3.637 ms 220-128-8-53.tyfo-3031.hinet.net (220.128.8.53) 4.439 ms
 6 * 220-128-9-189.tylc-3336.hinet.net (220.128.9.189) 5.963 ms 220-128-8-17.tyfo-3335.hinet.net (220.128.8.17) 6.157 ms
 7 142.250.169.120 (142.250.169.120) 6.656 ms 72.14.209.178 (72.14.209.178) 6.153 ms 142.250.169.120 (142.250.169.120) 6.682 ms
 8 220-128-8-189.tylc-3336.hinet.net (220.128.8.189) 6.382 ms * 192.178.106.175 (192.178.106.175) 6.217 ms
 9 142.251.226.171 (142.251.226.171) 5.460 ms 72.14.209.178 (72.14.209.178) 6.405 ms 72.14.213.90 (72.14.213.90) 6.324 ms
10 dns.google (8.8.8.8) 5.708 ms 6.075 ms 6.927 ms

```